Global Value Chains in Extractive Industries

A Case Study of Mining, Labour Precarity, and Value Capture in Odisha

AUTHOR
Ananya Pradhan

AFFILIATION
Centre for Informal Sector and Labour Studies (CISLS), School of Social Sciences (SSS), Jawaharlal Nehru University (JNU)

ABSTRACT
This paper investigates the political economy of Global Value Chains (GVCs) and Global Production Networks (GPNs) within the extractive sector, focusing on bauxite mining and aluminium refining in Odisha, India. Drawing upon theoretical frameworks of value capture, multi-tiered labour subcontracting, and labour control regimes, the study evaluates the impact of transnational capital on local labour processes and livelihood security through an empirical case study of Vedanta Limited's operations in Lanjigarh and surrounding mining clusters. Based on primary fieldwork comprising 35 semi-structured interviews and 5 Focus Group Discussions (FGDs) alongside longitudinal data from the Indian Bureau of Mines (IBM) and official state reports, the findings demonstrate that while mineral extraction integrates the regional economy into global commodity markets, it generates severe socio-economic vulnerabilities. Transnational extraction drives agrarian dispossession and environmental degradation, creating a localized 'reserve army of labour.' Rather than fostering formal, secure employment, lead firms and their intermediaries utilize multi-layered contract systems to enforce extended work shifts, sub-minimum wages, and unsafe working conditions, while suppressing collective bargaining and trade unionism with the aid of the state apparatus. The paper concludes that extractive GVC integration deepens labour precarity and institutionalizes structural inequalities between transnational capital and local working classes.

KEYWORDS
Global Value Chains, Global Production Networks, Extractive Industries, Bauxite Mining, Informal Labour, Labour Precarity, Odisha, Vedanta, Indian Bureau of Mines

1 Introduction

The integration of geographically dispersed regions alongside the functional fragmentation of the production process of a single commodity is conceptualized in contemporary political economy as a Global Production Network (GPN) or Global Value Chain (GVC). The ascendance of transnational corporations (TNCs) has led to substantial spatial reconfigurations in production and demand. In recent decades, the number of private corporations operating in extractive resources—a domain historically dominated by state-owned enterprises—has increased markedly due to the aggressive penetration of global capital.

This paper investigates the impacts of the extension of Global Production Networks in the extractive sector upon the workers employed within them, taking bauxite mining and aluminium refining in Odisha by the multinational enterprise Vedanta Resources as an illustrative case study. This inquiry critically engages with claims made by several GPN researchers that integrating domestic resource frontiers into global extractive circuits inherently drives regional development (Bridge 2008). In contrast, this paper argues that while the influx of transnational capital into extractive industries accelerates resource outflow and enclave growth, the employment generated remains deeply volatile, informalized, and precarious. The systematic reliance on third-party intermediaries, such as labour contractors, constitutes a primary institutional mechanism driving this precarity. Accordingly, the objective of this paper is to examine the contribution of GPN integration to the economic conditions of workers, the uncertainties governing their employment, and the resulting erosion of their collective bargaining power from a critical political economy perspective.

1.1 Meaning and Operation of Global Value Chains

The Global Value Chain (GVC) paradigm has emerged as a dominant framework for analyzing contemporary international trade and industrial organization. Although cross-border production activities have historical antecedents, value chain fragmentation gained unprecedented momentum following worldwide economic liberalization and globalization. Whereas traditional international trade was conceptualized primarily on the basis of comparative advantage in finished goods between nation-states, technological revolutions, logistical innovations, and institutional deregulations have enabled the functional unbundling of production across multiple countries.

A global value chain encompasses the full range of activities required to bring a product from its initial conception and raw material extraction, through intermediate stages of manufacturing and processing, to final distribution, marketing, and post-consumer disposal—with value added at each discrete stage. Lead firms strategically locate segments of production in geographical territories that maximize returns on capital investment. Consequently, global trade is characterized by the spatial integration of discrete territories accompanied by the functional disintegration of production processes. Today, intra-firm and inter-firm trade coordinated through cross-border production networks accounts for approximately 80 percent of global trade. The driving motivations behind this spatial fragmentation include heightened international competition among producers, access to cheap and disciplined labour, proximity to natural resources, and regulatory arbitrage across differing geographical regimes.

A value chain perspective reveals how production, distribution, and consumption are structurally mediated by the social relations of production governing input acquisition, manufacturing, and marketing. Broadly, global value chains represent the full sequence of coordinated functional activities required in value creation across national borders, wherein transnational corporations act as lead coordinating entities.

Governance structures within value chains are broadly differentiated into “producer-driven” and “buyer-driven” commodity chains (Gereffi 1994; Gereffi et al. 2005). Producer-driven chains are coordinated by large, capital- and technology-intensive transnational manufacturers that control critical production technologies and backward-forward linkages, typical in sectors such as automobiles, aircraft, and heavy machinery. Buyer-driven chains, by contrast, are governed by large retailers, branded merchandisers, and trading companies that coordinate decentralized, labour-intensive manufacturing networks across developing nations in consumer goods like garments, footwear, and toys. In these networks, lead firms retain high-value, rent-generating functions such as design, research and development, and brand marketing, while outsourcing lower-margin, labour-intensive manufacturing tasks to competitive tiers of global suppliers.

In developing economies, value chain integration has expanded across all sectors, including traditional extractive activities such as agriculture and mining. Extractive operations involve the systematic removal and primary processing of minerals, metals, and fossil fuels from the earth. Under contemporary GVC integration, extractive activities have been restructured through export-oriented, capital-intensive enclaves managed by transnational corporations. In mineral-rich developing countries like India, mining occupies a strategic position in national industrialization and foreign exchange generation. Because mining provides indispensable primary feedstock for heavy industry and global supply chains, it attracts massive inflows of transnational capital seeking access to high-grade mineral reserves.

Odisha holds an exceptionally prominent position in India’s mineral economy, ranking first nationally in the production of chromite (accounting for 87.93% of national output) and manganese ore (36.39%), and fourth in iron ore production. Crucially, the state contains massive deposits of high-grade bauxite, the primary raw material for alumina refining and aluminium smelting. Following the adoption of economic liberalization policies in 1991, the Indian state increasingly opened mineral extraction to private and multinational corporations through long-term mining concessions and leases. Transnational conglomerates, most notably Vedanta Resources—operating through subsidiaries such as Vedanta Aluminium Limited at Lanjigarh (alumina refinery) and Jharsuguda (smelter and power plant)—alongside corporate groups like Tata, Aditya Birla/Hindalco, and Jindal, established expansive operations integrated into international commodity circuits. Vedanta Resources maintains corporate headquarters in London with extensive mining operations spanning India (Odisha, Goa, Tamil Nadu, Punjab), Australia, and Zambia.

The Mines Act of 1952 provides the statutory definition of a mine as any excavation undertaken for searching or obtaining minerals, whether open-cast or underground, encompassing all associated operational processes (Ministry of Labour and Employment, Government of India 1952). The Act legally defines an owner as the immediate proprietor or occupier of a mine. Within this statutory framework, mine labour constitutes the essential human factor upon which mineral exploitation and downstream value creation depend. However, while resource extraction generates substantial export revenues and corporate profits, it imposes severe environmental and cultural displacement upon local communities. This contradiction raises urgent questions regarding the working conditions, occupational safety, social security, and fundamental rights of labour engaged at the base of the global extractive chain.

1.2 Background of the Case Study: Vedanta Aluminium Limited in Odisha

To ground these theoretical dynamics empirically, this study focuses on the labour process and socio-economic conditions of workers engaged by Vedanta Limited (VL), a major subsidiary of Vedanta Resources, in Odisha. Vedanta Limited is one of India's foremost producers of primary aluminium and metallurgical grade alumina, supplying key inputs to diverse industrial sectors including power, construction, defence, ceramics, and automotive manufacturing. Operating under the commercial brand name "VEDANTAL", the company produces aluminium ingots (comprising 75% of its core portfolio), billets (13%), and wire rods (12%). The company supplies both domestic industrial markets through nationwide distribution hubs and exports approximately 60 percent of its total output to global clients across South Asia, the Middle East, and East Asia.

Vedanta's operational complex in Odisha centers upon a 1-million-tonne-per-annum alumina refinery and captive power plant at Lanjigarh in Kalahandi district, integrated with smelter operations in Jharsuguda. To feed its refining operations, the company extracts and procures bauxite ore from regional deposits in southwestern Odisha, including the Karlapat and Kodingamali mines located in the bauxite-rich Niyamgiri hill tract. The operations of this corporate complex provide a critical site for investigating how global value chain integration shapes labour precarity, contract work systems, and community livelihoods at the extractive frontier.

2 Literature Review

A substantial body of interdisciplinary scholarship examines the organizational architecture, power dynamics, and labour implications of Global Value Chains and Global Production Networks. This review synthesizes key conceptual debates directly relevant to extractive industries and labour precarity.

Contemporary capitalism is fundamentally characterized by the functional disaggregation of production and consumption across national jurisdictions under the organizational coordination of networked transnational corporations (Gereffi and Korzeniewicz 1994). To analyze this structure, Gereffi (1994) conceptualizes Global Commodity Chains (GCC) as inter-organizational networks centered on specific commodities, functionally linking households, enterprises, and nation-states within the world economy. Drawing upon Hopkins and Wallerstein (1986, 159) foundational definition of a commodity chain as "a network of labour and production processes whose end result is a finished commodity," this approach demonstrates how value creation, distribution, and surplus appropriation are shaped by underlying social relations of production.

Expanding upon this tradition, the Global Production Network (GPN) framework conceptualizes economic globalization as an integrated, multi-scalar network of relational power and institutional embeddedness (Henderson et al. 2002). Henderson et al. argue that GPN analysis provides a critical lens for understanding spatial economic integration and the structural asymmetries of economic and social development. Regional transformation occurs through dynamic flows of capital, technology, and labour, with profound implications for the distribution of corporate power, value creation, and value capture across distinct geographic nodes.

However, this globalized restructuring has profound, and frequently adverse, ramifications for the working class. The hyper-competitive pressures governing transnational firms in GPNs drive relentless cost-minimization strategies, flexibility imperatives, and labour process intensification (Jha and Chakraborty 2014, 2016). Lead firms systematically exploit regional wage differentials and weaken standard employment relationships to lower production costs and extract maximum surplus value.

In her seminal work on labour in GPNs, Stephanie Barrientos (2013) highlights how global sourcing increasingly relies upon commercialized labour contracting and third-party intermediaries. While value chain expansion ostensibly offers wage-earning opportunities to asset-poor populations, the widespread insertion of labour contractors institutionalizes precarious and "unfree" labour regimes. Analyzing global agricultural and horticultural chains, Barrientos illustrates how a "cascade system" of labour subcontracting allows lead firms to externalize risks and liability onto vulnerable workers, fostering conditions of debt bondage and extreme exploitation. Similarly, empirical investigations in the Indian context demonstrate that multi-tiered contract labour systems enable employers to depress wages, heighten numerical flexibility, offload supervisory burdens, and effectively dismantle collective bargaining and trade union rights (Barnes et al. 2015; Annavajhula and Pratap 2012).

The institutional mechanisms disciplining labour in global supply chains are systematically analyzed by Mark Anner (2015) through the framework of "labour control regimes." Anner identifies three interrelated regimes of control:

1. *State control*, wherein legal, administrative, and coercive state machinery curtails independent unionization and collective action;
2. *Market despotism*, wherein persistent unemployment, economic vulnerability, and the existential fear of job loss discipline workers; and
3. *Employer repression*, encompassing overt coercion, intimidation, and violence against union organizers.

These control regimes frequently operate in tandem across developing host economies to prevent production disruptions and ensure uninhibited capital accumulation.

In the post-liberalization period, Odisha emerged as a primary destination for domestic and multinational capital investments in large-scale, capital-intensive mineral projects. State developmental policy has actively facilitated resource extraction on the premise of overcoming chronic regional poverty, underemployment, and agrarian stagnation (Mishra 2010). However, critical scholars emphasize that the socio-ecological costs of this extractive model fall overwhelmingly upon local Adivasi and Dalit populations (Ambaguda 2010; Saxena et al. 2010; Padel et al. 2010; Kumar 2014; Debranjan 2008). Large-scale bauxite mining involves extensive deforestation, blasting, topsoil removal, and water table depletion, which directly undermine customary subsistence agriculture and forest gathering. Rather than delivering broad-based regional prosperity, extractive enclaves generate extensive displacement, environmental destruction, and minimal secure employment for local inhabitants, provoking sustained socio-political conflict between indigenous communities, transnational corporations, and the state apparatus.

3 Statement of the Problem and Research Gap

While extensive scholarly literature has analyzed the macroeconomic dimensions of transnational capital flows and value chain governance in manufacturing and agro-food sectors, critical empirical research on the labour process and worker precarity in extractive GVCs remains limited. The existing literature presents two sharply contrasting narratives: on the one hand, mainstream GPN perspectives posit that integrating resource-rich regions into global production circuits stimulates regional growth, employment generation, and socio-economic upgrading (Bridge 2008); on the other hand, critical labour scholars argue that global supply chains institutionalize informalization, multi-layered subcontracting, and the suppression of collective bargaining to maintain global cost competitiveness (Barrientos 2013; Jha and Chakraborty 2016).

This study addresses this research gap by conducting an empirical examination of bauxite mining and alumina refining in southwestern Odisha operated by Vedanta Limited. By investigating the specific mechanisms of labour recruitment, wage determination, workplace safety, and union suppression, the paper provides grounded insights into how extractive GVC integration reshapes labour-capital relations in resource-rich but economically marginalized regions.

4 Objectives of the Study

The specific research objectives of this paper are:

1. To examine the problems faced by mineworkers and refinery workers regarding workplace conditions, wage remuneration, occupational safety, and bargaining power within an extractive Global Production Network.
2. To analyze the impacts of GVC-led extractive operations on the socio-economic status, livelihoods, and living standards of local working-class and indigenous communities.

5 Methodology and Research Design

To investigate the labour process and socio-economic dynamics in Odisha’s extractive corridor, this study employs a mixed-method case study design combining primary qualitative field research with secondary longitudinal and policy data analysis.

5.1 Primary Fieldwork and Sampling Strategy

Primary fieldwork was conducted across the Lanjigarh block in Kalahandi district and contiguous mining villages in Rayagada and Koraput districts. Due to heightened corporate security gatekeeping, intense surveillance, and the complete absence of recognized trade unions within the industrial belt, formal random sampling was unfeasible. Accordingly, respondents were selected through non-probability purposive and snowball sampling strategies, utilizing initial entry points facilitated by local civil society organizers, community elders, and experienced informal worker networks.

Primary qualitative data collection comprised two principal research instruments:

1. **Semi-Structured In-Depth Interviews ($N = 35$):** Conducted individually with contractual mineworkers ($n = 12$), refinery contract labourers ($n = 10$), displaced Adivasi (Dongria and Kutia Kondh) and Dalit villagers ($n = 8$), local labour intermediaries/sardars ($n = 3$), and trade unionists/human rights activists ($n = 2$). Interviews focused on daily shift lengths, overtime compensation, wage structures, recruitment tiers, safety provisions, and grievances regarding workplace discipline.
2. **Focus Group Discussions (FGDs, $K = 5$):** Five focused group discussions were organized, comprising 5 to 6 participants per session (totaling 27 participants). Three FGDs were conducted with mixed-tier contract mineworkers and refinery helpers to capture collective perceptions of job insecurity, employer surveillance, and lack of unionization. Two gender-inclusive community FGDs were held with displaced villagers to examine agrarian displacement, red-mud environmental contamination, and loss of traditional forest-based livelihoods.

Table 1 summarizes the profile of primary research participants and methodological tools.

Table 1: Primary Fieldwork Sampling and Informant Profile

Respondent / Group Category	Research Tool	Sample Size / Composition	Core Focus of Inquiry
Contractual Mineworkers	Semi-Structured Interviews	$N = 12$	Daily wages (Rs. 300–350), 8+ hour shifts, lack of safety gear, sub-contracting tiers.
Refinery Floor Contract Labourers	Semi-Structured Interviews	$N = 10$	Overtime non-payment, PF deductions (Rs. 650–900), health hazards, supervisor co-optation.
Displaced Adivasi & Dalit Villagers	Semi-Structured Interviews	$N = 8$	Land alienation, red-mud waste pollution, decline in crop yields, water depletion.

Respondent / Group Category	Research Tool	Sample Size / Composition	Core Focus of Inquiry
Labour Intermediaries (Sardars)	Key Informant Interviews	$N = 3$	Recruitment mechanisms, commission systems, labour control for OSL.
Trade Unionists & Civil Rights Activists	Key Informant Interviews	$N = 2$	Criminalization of labour unrest, state-corporate nexus, barriers to union registration.
Mine & Refinery Contract Labourers	Focus Group Discussions	3 FGDs ($n = 16$)	Collective precarity, fear of termination/blacklisting, failure of company welfare committees.
Displaced Farmers & Women Villagers	Focus Group Discussions	2 FGDs ($n = 11$)	Collapse of customary *Dongar* cultivation, loss of forest produce, food insecurity.
Total Primary Research Sample	Mixed Qualitative Methods	**35 Interviews + 5 FGDs (62 Total Informants)**	Comprehensive empirical mapping of the extractive corridor.

5.2 Secondary Data and Statutory Review

To situate the primary micro-level findings within broader macro-structural trends, secondary data were systematically collected and analyzed from the following institutional sources:

1. **Indian Bureau of Mines (IBM):** Longitudinal state-level data on bauxite mineral production, mine reporting, and average daily employment from successive volumes of the *Indian Minerals Yearbook* (Indian Bureau of Mines (IBM) 2017).
2. **Directorate of Mines, Government of Odisha:** Administrative reports on mineral concessions, royalty generation, and district-level mineral output (Department of Steel and Mines, Government of Odisha 2018).
3. **Statutory and Policy Documents:** Provisions of the Mines Act (Ministry of Labour and Employment, Government of India 1952), the Contract Labour (Regulation and Abolition) Act (Ministry of Labour and Employment, Government of India 1970), and the Report of the Four-Member Committee on Bauxite Mining in Niyamgiri (Saxena et al. 2010).

5.3 Methodological Limitations and Ethical Safeguards

Conducting fieldwork in an extractive industrial enclave characterized by intense social tensions and corporate-state surveillance posed acute methodological challenges. Because contract workers face immediate dismissal and blacklisting for speaking critically of company management, absolute anonymity was guaranteed to all participants. Names, specific contractor affiliations, and village-level identifiers have been pseudonymized to ensure ethical integrity and protect respondents from occupational or physical retaliation.

6 Analysis and Empirical Findings

6.1 Macro-Level Trends: Capital Intensity and Jobless Growth in Bauxite Extraction

The mining and quarrying sector in Odisha has undergone extensive restructuring following the 1991 macroeconomic reforms and the 2001 disinvestment drives. While the sector was historically presented as a catalyst for mass regional employment, contemporary mineral extraction is characterized by high capital intensity and labour-displacing mechanization.

Longitudinal data compiled from the Indian Bureau of Mines (IBM) and the Directorate of Mines, Government of Odisha, reveal a striking divergence between soaring mineral extraction volumes and stagnant direct employment levels in bauxite mining across the state, as detailed in Table 2.

Table 2: Bauxite Production and Average Daily Direct Employment in Odisha (2012–13 to 2016–17)

Financial Year	Bauxite Production (Million Tonnes)	Share in All-India Production (%)	Average Daily Direct Employment (Persons)
2012–13	6.88	41.6%	895
2013–14	7.95	36.7%	674
2014–15	8.36	37.6%	674
2015–16	10.87	40.8%	896
2016–17	13.89	56.1%	1,215

Source: Compiled by author from Indian Bureau of Mines (IBM), Indian Minerals Yearbook (various issues), and Directorate of Mines, Government of Odisha.

The data in Table 2 illustrate the pronounced capital-intensive nature of extractive value chains. Between 2012–13 and 2016–17, bauxite production in Odisha expanded dramatically from 6.88 million tonnes to 13.89 million tonnes—a massive increase of 101.8 percent—propelling Odisha's share to 56.1 percent of total national output. Yet, recorded direct daily employment grew by only 35.7 percent, fluctuating between a low of 674 workers and a peak of 1,215 workers for the entire state. This stark asymmetry empirically confirms that globalized mineral extraction is fundamentally characterized by "jobless growth" in formal employment, forcing the regional labour surplus into unrecorded, informal contractual arrangements.

6.2 Corporate Architecture, Subcontracting Cascades, and the Intermediary Regime

Vedanta Aluminium Limited—headquartered in Jharsuguda, with equity held between Vedanta Resources (70.5%) and Sterlite (29.5%)—operates an expansive complex comprising the Lanjigarh alumina refinery and regional bauxite procurement operations in Karlapat and Kodingamali. To minimize direct wage liabilities and circumvent statutory responsibilities under the Mines Act and Contract Labour Act, the lead firm relies on an elaborate, multi-layered subcontracting cascade.

Primary handling, logistical, and extraction contracts are awarded to major intermediary firms, notably Odisha Stevedores Limited (OSL). In turn, OSL delegates operational tasks to secondary and tertiary local labour contractors (*sardars*). Primary interviews with intermediaries reveal that these labour contractors are deliberately selected from individuals wielding local political patronage and caste influence. This intermediary cascade allows the lead multinational firm to insulate itself from legal accountability, transfer market fluctuations onto labour, and depress compensation across the value chain.

6.3 Agrarian Dispossession, Ecological Degradation, and the “Reserve Army of Labour”

Prior to the entry of large-scale industrial extraction, local Adivasi communities (Dongria Kondh, Kutia Kondh, Pengo) and resident Dalit households maintained diversified agro-ecological livelihood systems. Mountain plateau dwellers practiced customary shifting cultivation (*dongar chas*)—producing drought-resilient crops of upland paddy, millets, mustard, and pulses—supplemented by cattle rearing and minor forest produce collection. Valley inhabitants engaged in settled wet agriculture.

Fieldwork narratives collected during individual interviews and community FGDs indicate that the expansion of the industrial complex severely disrupted this subsistence economy. Extensive land acquisition, heavy machinery movement, and widespread deforestation alienated local cultivators from customary grazing and agricultural tracts. Concurrently, refinery operations generated acute environmental contamination: the discharge of caustic red mud slurry and airborne particulate matter severely polluted local streams, lowered the water table, and drastically degraded adjacent agricultural soils. Consequently, agricultural yields suffered a catastrophic collapse.

In political economy terms, this process represents classic accumulation by dispossession. By degrading the ecological foundations of customary agriculture, the corporate complex systematically detached local producers from their means of subsistence. Stripped of viable agrarian alternatives, local populations were transformed into an impoverished, highly vulnerable “reserve army of labour” compelled to accept precarious, sub-minimum wage employment within the very extractive apparatus that dispossessed them.

6.4 Workplace Regimes: Shift Violations, Stagnant Wages, and Social Security Exclusion

Primary interviews with mineworkers and refinery labourers ($N = 22$) provide detailed evidence of exploitative workplace conditions. Operations function continuously on a three-shift rotation across 24 hours. Although statutory regulations legally mandate an 8-hour workday, respondents consistently testified that contract workers are routinely forced to perform 10 to 12 hours of continuous manual labour per shift, while receiving payment strictly for standard 8-hour daily rates without overtime compensation.

Daily wage rates for contractual workers range between Rs. 300 and Rs. 350 for general manual and hazardous operations, falling significantly below statutory recommendations for skilled industrial work in hazardous zones. Furthermore, sharp institutional segmentation governs social security: while a minor subset of refinery shopfloor workers receives modest Provident Fund (PF) allocations (ranging between Rs. 650 and Rs. 900 per month), contractual workers engaged in open-cast excavation and ore handling receive zero statutory PF, pension, or health insurance coverage. On-site healthcare is restricted to elementary first-aid facilities.

The enterprise maintains a rigid division of labour: permanent managerial and supervisory positions are awarded almost exclusively to non-local, highly paid technical staff recruited from outside the region, whereas arduous, life-threatening tasks—such as bauxite crushing, chemical slurry transfer, and pit excavation—are allocated entirely to local Adivasi, Dalit, and migrant contract labourers.

6.5 Suppression of Collective Action, Workforce Segmentation, and the State-Capital Nexus

Autonomous trade unionism is entirely prohibited within the industrial complex. Focus group participants emphasized that any attempt to organize independent worker associations or demand compliance with statutory safety norms is met with immediate dismissal, wage withholding, and systematic blacklisting across regional contractor networks. To maintain an appearance of worker representation, the enterprise sponsors internal welfare committees that possess zero independent bargaining capacity.

This structure aligns directly with Jha and Chakraborty (2014) and Jha and Chakraborty (2016) analysis of post-Fordist labour segmentation. By creating a deep structural cleavage between a co-opted permanent supervisory core and a disenfranchised contractual periphery, the enterprise achieves absolute numerical and wage flexibility. Production targets and market volatility are managed by unilaterally intensifying workloads during high-demand cycles and executing abrupt, uncompensated layoffs during market slumps.

When workers and affected villagers organize demonstrations or public grievances, the corporate administration mobilizes the coercive apparatus of the local state. Protesting workers and civil rights organizers are routinely branded as “anti-development” and “anti-state”, facing arbitrary detention and police intimidation. As Marx and Engels famously observed in the *Communist Manifesto* (1848):

> “The executive of the modern state is but a committee for managing the common affairs of the whole bourgeoisie.”

In the extractive political economy of Odisha, state institutions function not as neutral regulators of statutory labor law, but as the active coercive guarantor of transnational corporate accumulation, securing resource corridors and disciplining labour to guarantee uninterrupted capital flows.

7 Conclusion

The mainstream developmental thesis that foreign capital investment in mineral extraction inherently promotes regional development and socio-economic upgrading is fundamentally refuted by the empirical reality of Odisha's bauxite mining corridor. Instead, the incorporation of the regional economy into the global aluminium production network reflects the classic architecture of an extractive enclave. Transnational capital captures substantial economic rents through the global marketing of high-value aluminium products, while externalizing ecological devastation, agrarian collapse, and acute labour precarity onto local working-class and indigenous communities.

This study has shown that integration into extractive GVCs deepens labour vulnerability through structural mechanisms: the systematic degradation of agrarian livelihoods generates a localized reserve army of labour; a multi-tiered subcontracting hierarchy insulates lead firms from statutory liabilities; work shifts are unilaterally prolonged without overtime pay; and autonomous trade unionism is suppressed through the joint coercion of corporate management and the state apparatus. Consequently, extractive GVC integration does not foster human development; rather, it institutionalizes a deeply unequal structural relationship between global capital and peripheral labour at the resource frontier.

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